

STAREHE GIRLS' 2025 PREDICTION



POSSIBLE KCSE EXAM



443/1

AGRICULTURE

PAPER 1 (THEORY)

TIME: 2 HOURS

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

(Kenya Certificate of Secondary Education)

Instructions to students

- a) Write your name and Admission number in the spaces provided at the top of this page*
- b) This paper consists of three sections: A , B and C*
- c) Answer all questions in sections A and B*
- d) Answer any two questions in section C*
- e) All answers should be written in the spaces provided*
- f) Candidate should answer the questions in English*

SECTION A (30 Marks)

Answer all questions in this section on the spaces provided

1. State **four** factors considered when identifying a farming system (2 marks)

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2. What is training as used in crop production (1 marks)

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3. Name **two** types of planting materials. (1 mark)

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4. Give **two** ways in which soil structure influences crop production . (1 mark)

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5. State **four** importance of raising seedlings in a nursery bed. (2 marks)

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6. State **one** importance of soil colour in crop production. (1 mark)

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7. Mention **three** methods of primary cultivation (1½ marks)

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8. Give **four** reasons why green manures are not commonly used by farmer to enrich the soil (2 marks)

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9. a) State **four** factors considered when designing a crop rotational programme. (2 marks)

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b) Give **two** reasons why maize should not be alternated with rice in a rotational programme.

(1 mark)

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10. State **four** importance of organic matter in the soil .

(2 mark)

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11. a) What is seed scarification .

(1 mark)

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b) Give **three** ways of achieving seed scarification.

(1½ marks)

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12. Give **three** types of information contained in a labour record (1½ marks)

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13. a) What is “government policy” as used in agriculture (1 mark)

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b) Give three ways in which the government regulates the production and sale of agricultural goods.
(1½ marks)

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14. State **three** ways of modifying the soil pH . (1½ marks)

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15. State **two** effects of bacteria on crop production (1 mark)

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16. Differentiate between rolling and ridging as used in crop production. (1 mark)

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17. Name **three** poor farming practices that lead to water pollution in the farm. (1½ marks)

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18. What is “topping” as used in forage production (1 mark)

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19. List **two** ways through which water is collected and stored in most farms (1 mark)

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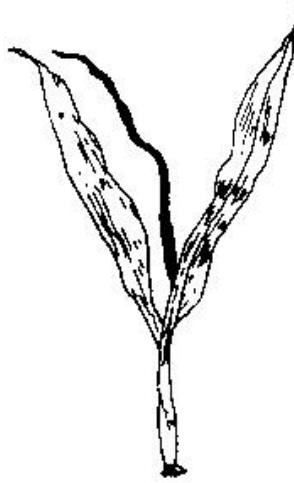
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SECTION B (20 marks)

Answer all questions in this section on the spaces provided

20. Below is a diagram of sorghum seedling that has been attacked by a certain pest



a) Identify the pest

(1 mark)

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b) Identify the developmental stage of the pest named in (a) above that causes the damage.

(1 mark)

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c) State the damage that has been caused by the pest named in (a) above. **(1 mark)**

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d) Give **two** control measures that can be used to control the pest named in (a) above **.(2 marks)**

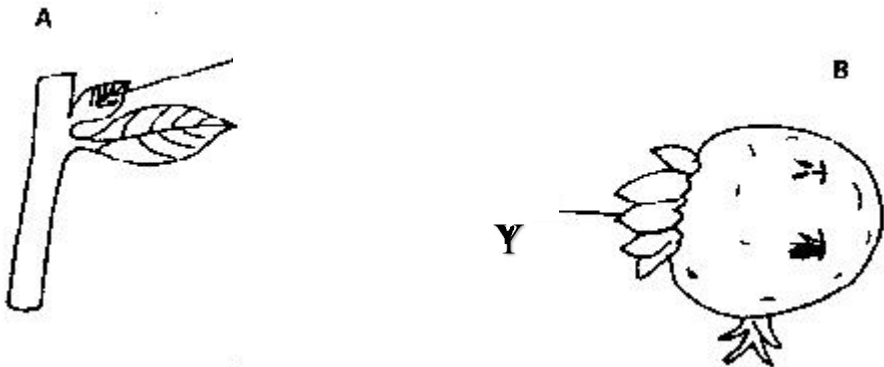
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21. The illustration below shows materials used in vegetative propagation, labelled A and B



a) i) Identify the vegetative materials A and B

A (½ mark)
 B..... (½ mark)

ii) Name the parts labelled X and Y

X (½ mark)
 Y(½ mark)

iii) Name the preparation method used to get the vegetative material B (1 mark)

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iv) State two disadvantages of using vegetative materials for planting (2marks)

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22. The illustration below shows a tomato fruit affected by a disease or a condition



a) Identify the condition illustrated

(1 mark)

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b) State **three** conditions that predispose a tomato plant to the condition stated in (a) above

(3 marks)

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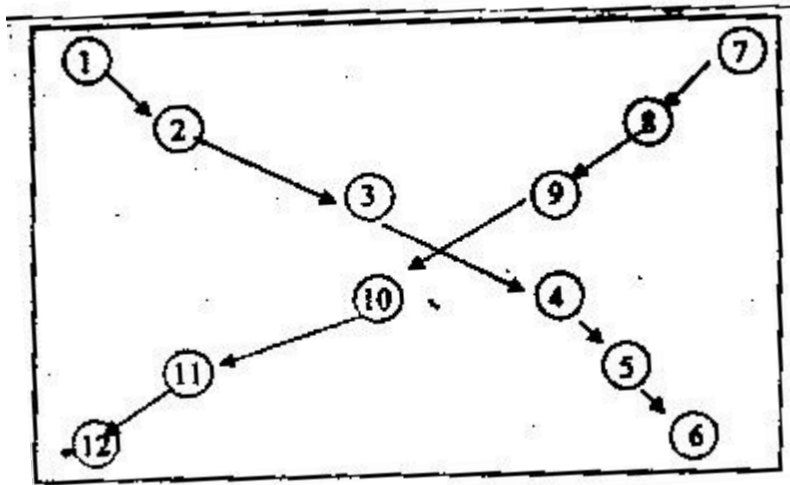
c) Name **two** other diseases that affect tomatoes

(1 mark)

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23. Below is an illustration showing a soil sampling method. Use it to answer the questions that follow.



a) What method of soil sampling is represented above (1 mark)

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b) Outline **two** areas which should be avoided during soil sampling (2 marks)

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c) State the **two** sets of information that should accompany the composite soil sample to the laboratory for analysis. (2 marks)

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SECTION C (40 MARKS)

Answer any two questions in this section in the spaces provided after question 26.

24. a) Explain the advantages of grass-legume pastures over a pure grass pasture. (8 marks)
- b. Describe the effects of fragmentation and sub-division of land in our country. (6 marks)
- c) Explain **six** biotic factors that influence agricultural production. (6 marks)

25. a) Explain the process of treating water in the farm using the chemical treatment system.

(12 marks)

b) Explain **six** physical factors that increases the rate of soil erosion in farm. **(6 marks)**

c) Explain how monoculture leads to loss of soil nutrients . **(2 marks)**

26. a) Explain **ten** precautions that a farmer should observe when using farm chemicals .**(10 mks)**

b) Give **five** reasons for pruning in coffee production. **(5 marks)**

c) Describe **five** factors that influence soil formation . **(5 marks)**

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